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Physics 0525: Lab 8

Op-amps: negative feedback

Equipment: Breadboard, DMM, signal generator, oscilloscope, LF411, 10K Ω pot

Preliminaries:

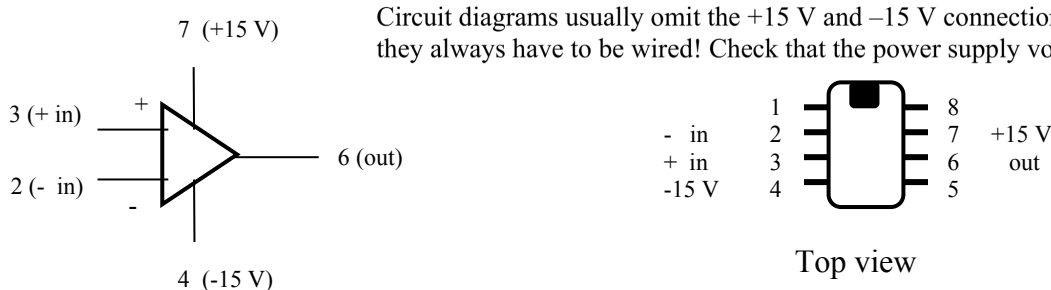
Operational amplifiers (op amps) are integrated circuits (IC) containing a bunch of transistors, resistors, and capacitors. They have two input pins, one inverting (-) and one non-inverting (+), a single output pin, plus power pins and some other. An op amp is a differential amplifier, i.e. the output voltage is proportional to the difference between the input voltages: $V_o = \beta(V_+ - V_-)$.

The so called open-loop voltage gain β is so high (typically $10^5 \sim 10^6$) that the output will swing to the high or low power supply voltage for even a tiny difference between the two inputs. For this reason, the op amp is practically never operated without negative feedback when a fraction of the output is coupled back to the inverting (-) input of the amplifier. As a result of this negative feedback, the amplifier gain becomes very stable and can be controlled precisely by the external components in the feedback circuit.

The operation of any op amp with negative feedback can be analyzed using the following rules of thumb:

- (1) The input impedance of both inputs is so high that the inputs practically draw no current.
- (2) The output of the op amp changes in a way to make the voltages at the two inputs practically equal ($V_+ = V_-$).

The figure below shows the schematic symbol and pin diagram of a typical op amp housed in a 8-lead dual in-line package (DIP). There is a dot in the corner or a notch at one end of the chip that identifies the end from which to begin counting the pins (in counterclockwise direction).



Circuit diagrams usually omit the +15 V and -15 V connections to the power supply but they always have to be wired! Check that the power supply voltages are set to $\pm 15V$.

Lab 8: Op-amps w/ negative feedback

LF411

1 0 8
in 2 - 7 +15V
in 3 + 6 out
4 -15V 5

potentiometer (variable R)
+15V
10K-R
10K 2
R
3 -15V

① $i_+ = i_- = 0$ ② $V_+ = V_-$

① open-loop test ② Inverting amp ③ Non-inverting amplifier + Follower (V_+)

⑤ Summing amplifier

Gain-Bandwidth product

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Part 1: Open-loop test circuit

This first experiment illustrates the very high open-loop gain of the op amp. Set it up and use a small amplitude (symmetric) triangular wave (~500 mVpp, 1 kHz) from the signal generator as the input. Display both the input and output on the oscilloscope at the same time.

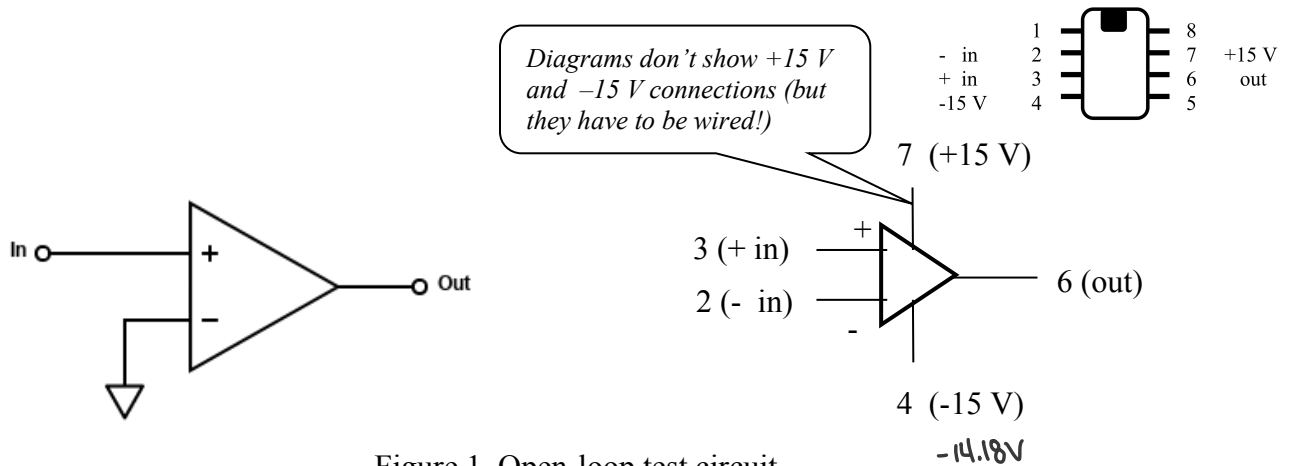
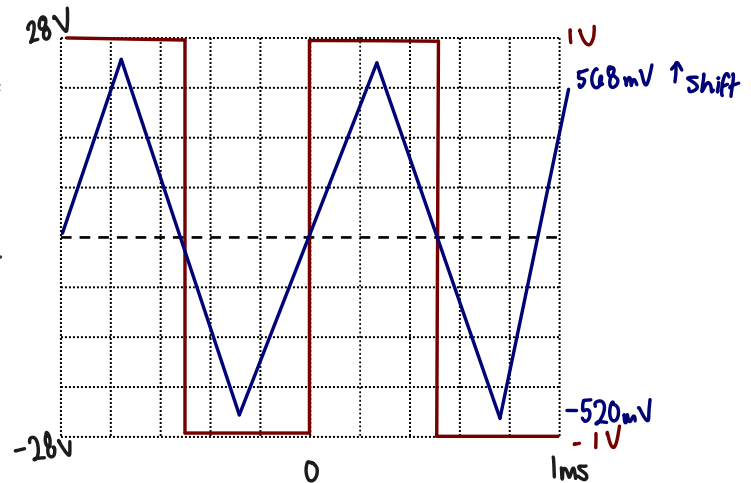


Figure 1. Open-loop test circuit.

Make a sketch of the input and output waveforms. What is the relationship between the output and the input? What does this tell you about the open-loop gain?

this tells us that A_{OL} is very large



Output saturation: The output of the op-amp is ultimately limited by the supply voltage ($\pm 15V$), but the actual limits are smaller due to the internal voltage drops inside the op-amp. Change the input to a small amplitude square wave (~500 mVpp, 1 kHz) and measure the positive and negative output saturation. Determine the voltage drop inside the op-amp (difference between the max/min output and the power supply voltages).

$$14.2 + 13 \quad / \quad 15 + 14$$

the voltage drop in the op-amp is about 202V

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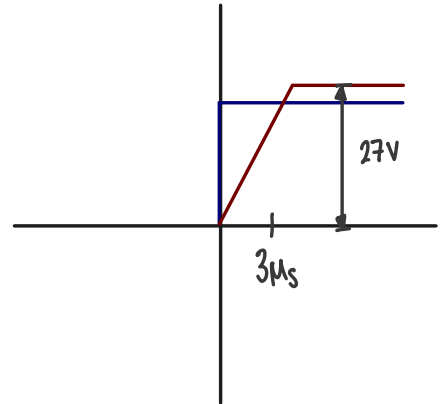
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Slew rate tells you how fast the output is rising or falling, usually this is given in V/ μ s. Use the same square wave input as above and zoom in on a rising (or falling) edge of the output. Increase the amplitude of the input until the slope of the edge in the output reaches its maximum. Use the cursor and measure the slope of the rising (or falling) edge. How does your measured slew rate compare to the value quoted in the op-amp data sheet?

$$\sim 27V / 3\mu s = 9V / \mu s$$

data sheet: min - $8V / \mu s$ typical - $15V / \mu s$

we are close to min!



Part 2: Inverting amplifier

The second circuit is much more useful than the first one thanks to the negative feedback. Assemble the circuit inside the dashed box (without the 1 k Ω input resistor). Apply a small amplitude 1 kHz sine wave and observe input and output on your oscilloscope at the same time.

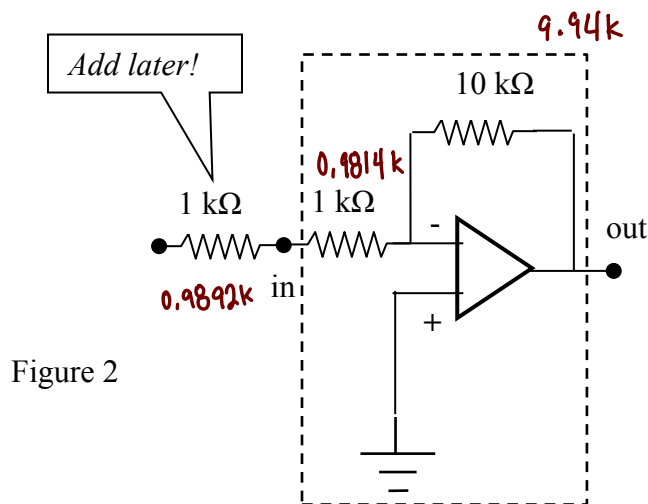


Figure 2

a) What is the phase relationship between input and output?

In phase []

out of phase by 180° [✓]

b) Measure the amplitude of the input and output waveforms. What is the voltage gain?

Measured voltage gain: ~ 9.915

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c) Calculate the expected voltage gain of the amplifier circuit:

$$I_{in} = \frac{V_{in}}{R_{i2}} = \frac{500\text{mV}}{1.9706\text{k}\Omega} = 253.73\text{A}$$
$$-\left(\frac{9.94}{0.9814}\right) = -10.1284$$

Expected voltage gain: -10.1284

d) Now add the 1 k Ω resistor in series with the input. What happens to the output voltage?

$$-\left(\frac{9.94}{1.9706}\right) = -5.044\text{V, the output is halved}$$

What does that tell you about the input impedance of the amplifier circuit (the circuit inside the dashed box of the figure)?

input impedance $\approx$ input resistor $\approx 1\text{k}\Omega$

Input impedance: -5.044 V [Ω]

e) Now try to measure the output impedance of the amplifier circuit by adding 1 k Ω load resistor connected to ground at the output of the circuit (no need for blocking capacitor). What is the amplitude of the output with and without the load?

the amplitude of the output does not appear to change

What does that tell you about the output impedance of the amplifier circuit?

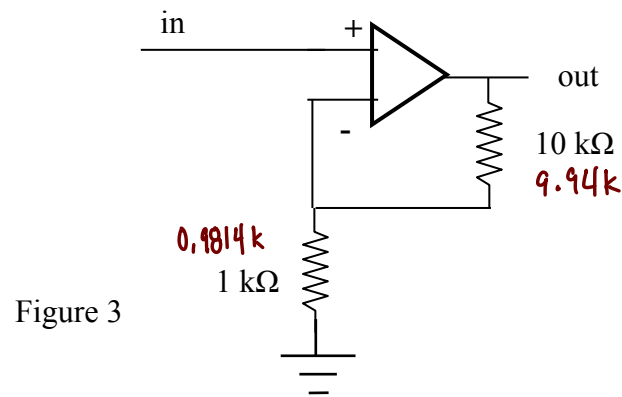
this tells us that the output impedance is much lower
than that of the input

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Part 3: Non-inverting amplifier

This circuit is very similar to the previous one (inside the dashed box) except that the input and ground connections are interchanged. It also has a similar (but not identical) gain. Test it with small amplitude sine wave.



a) What is the phase relationship between input and output?

In phase [☒] out of phase by 180° []

b) Measure the amplitude of the input and output, and determine the voltage gain of the amplifier:

Measured voltage gain: 11.05

c) Calculate the expected voltage gain of the amplifier:

$$V_{out} = \left(1 + \frac{9.941k}{0.9814k} \right) = 11.1294$$

Expected voltage gain: 11.1294

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d) Try to measure the circuit's input impedance by adding a $1\text{M}\Omega$ resistor in series with the input. Do you see any change in the amplitude of the output with and without the input resistor? What does that tell you about the input impedance?

no change, the addition of the $1\text{M}\Omega$ resistor increases the input's impedance

e) Try to measure the output impedance of the circuit by adding a $1\text{k}\Omega$ load resistor connected to ground at the output of the circuit. Does the amplitude of the output change with and without the load? What does that tell you about the output impedance of the circuit?

the addition of the $1\text{k}\Omega$ load resistor yields no change to the output of the circuit - telling us the output impedance is much lower than that of the input

Part 3b: High frequency cut off (or roll off)

Go back to the circuit on Figure 3 and change the feedback resistor from $10\text{K}\Omega$ to $100\text{K}\Omega$ so that the gain of the amplifier circuit is ~ 100 . Test the circuit using 1kHz symmetric triangular waves of 50mVpp amplitude from the signal generator.

Gradually increase the frequency of the input signal. Do you see any distortion in the output?

As we increase the frequency, the output starts to distort, losing its sharp triangular shape and closely resembling a sine-wave

Now repeat the test using sine waves by gradually increasing the frequency from 10kHz all the way to the maximum. What do you observe? Does this help to understand what happened with the triangular waves? Hint: the triangular wave can be considered as the sum of a series of sine waves with increasing frequency (Fourier theorem).

The amplitude drops while growing out of phase as the frequency increases.

Yes, this effect is due to the super

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Determine the -3dB cut-off frequency of the amplifier circuit? Do not compare the amplitude of the input to the output, rather determine when the amplitude of the output drops by the required fraction compared to the output measured at low frequency.

$$V_o = 10V$$

$$V = \frac{1}{\sqrt{2}} 10V \Rightarrow 7.071V$$

$$\omega \uparrow \Rightarrow f_{in} \approx 29 \text{ kHz}$$

Multiply the gain of the circuit with the 3dB cut-off frequency: this is called the gain-bandwidth product (GBW) of the op-amp? Compare with that in the data sheet.

$$2900 \text{ kHz} = 2.9 \text{ MHz}$$

$$\Rightarrow 2.7 - 4 \text{ MHz} \Rightarrow \text{within range}$$

Part 4: Voltage follower

It is very easy to build a voltage follower using an op amp. Note that this is a special case of the non-inverting amplifier circuit used in Part 3 with the feedback resistor set to zero and the other resistor (between - input and ground) set to infinite. Build the circuit and test it with sine waves at 1 kHz:

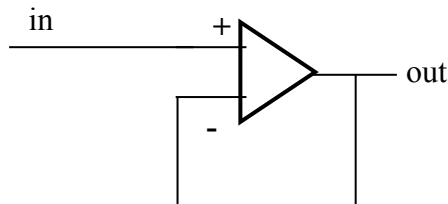


Figure 4

a) Measure the input and output wave amplitude. What is the voltage gain?

$$(1 + 0) = 1$$

b) Try measuring the input and output impedances as you did in Part 3. What do you conclude?

w/ 1M @ input: no effect observed in gain

- input impedance must be much larger

w/ 1k @ out : no effect observed in gain

& :
1M @ in - input impedance must be much smaller

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The circuit does not seem to do much. Why is it useful?

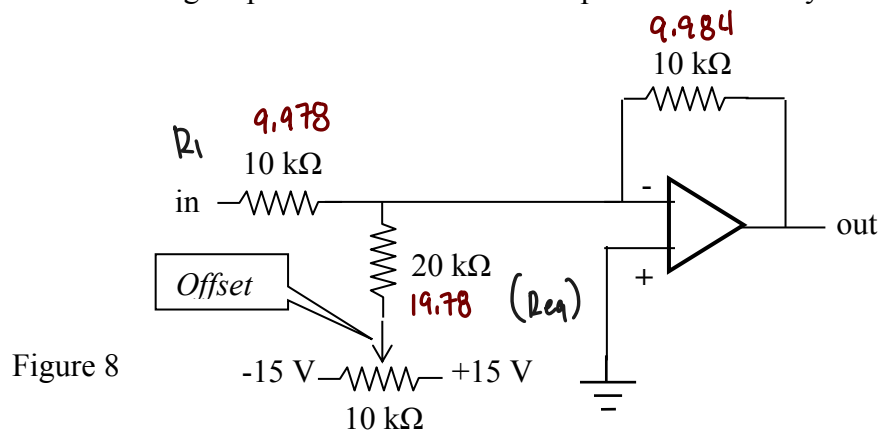
It allows us to measure absolute readings with the incorporated circuit elements impedances ignored.

V_{in} will match V_{out} , with the circuit acting as a buffer between input and output

Part 5: Summing amplifier

→ unity gain buffer

The summing amplifier is a modification of the inverting amplifier used in Part 2. The circuit below performs an analog addition of the signal fed into the input and the constant voltage set by the voltage divider pot. If you use a sine wave (or some other wave form) as the signal, the output will look like the signal plus a DC offset from the potentiometer. Try it out!



a) Write the formula that gives the output as a function of the two input signals from the function generator and the pot?

$$V_{out} = -\left(\frac{R_1}{R_{eq}}\right)(V_{in1} + V_{in2})$$

$$V_{out} = \frac{1}{R_F/R_1} \cdot V_{AC} - \frac{1}{2} V_{DC} \Rightarrow \frac{V_{AC}}{V_F/R_1} - \frac{1}{2} V_{DC}$$

b) Is the output consistent with the expectation above? What is the amplitude and DC offset of your amplifier output signal? What is the amplitude of the input from the function generator and the DC offset set by the pot (as measured on the scope)?

432 mV out
800 mV in

$$V_{DC\ off} = \frac{1}{2} V_{DC}$$

Done